

1. Why feasibility study of road projects is important and what are the main activities?

Answer:

A preliminary investigation conducted to determine whether a proposed road project is technically, economically, environmentally, and socially viable before proceeding to detailed design and construction is known as feasibility study. It is important because it:

- i. Helps to determine the viability of project by assessing whether the road project is economically and technically feasible and practical.
- ii. Helps in selecting the best alignment considering the technical, environmental and economic factors.
- iii. Provides cost estimates for the construction, operation and maintenance.
- iv. Identifies different technical challenges such as complex geography, unstable slopes, weak soils and drainage problems.
- v. Reduces the project risks by identifying potential hazards such as landslides, floods and seismic effects.
- vi. Improves the project planning and scheduling, minimizing delays and cost overruns.
- vii. Ensures sustainable development by balancing economic, social and environmental considerations.

The main activities included in feasibility study of road projects are:

- i. Reconnaissance survey:
All the necessary preliminary information is collected including the alternate road alignments, terrain, drainage, settlement, accessibility etc.
- ii. Topographical survey:
The ground features and contours are surveyed.
Maps, longitudinal sections, and cross-sections of road are to be prepared.
The suitable road alignment must be determined during this survey.
- iii. Traffic survey:
The ground features and contours are surveyed in traffic survey.
Traffic demand at present as well as future must be estimated.
Different type of vehicles are classified and design traffic is determined.
- iv. Geological and geotechnical investigation:
Soil and rock conditions are studied properly.
The landslide-prone areas must be identified.
The foundation conditions for bridges and retaining structures must be determined.
Slope stability must be properly assessed.
- v. Hydrological and drainage study:
Rainfall and runoff must be analyzed.
Design floods are estimated, soil erosion and scour risks are assessed.
Culverts, bridges and drainage structures must be planned for design.
- vi. Environmental and social assessment:
Impacts of the construction works on forests, wildlife, water bodies and air quality must be evaluated.
Land acquisition and resettlement requirements are assessed.
- vii. Economic and financial analysis:
Construction, maintenance and operation costs of road construction must be estimated.

Economic feasibility must be evaluated by conducting cost-benefit analysis.

- viii. Construction planning:
The availability of construction materials, labor and equipment are assessed.
Construction methodology and project schedule are prepared.
- ix. Risk assessment:
The landslides, floods, earthquakes and climate risks are identified.
Mitigation and disaster-resilient measures are recommended.
- x. Preparation of feasibility report:
All the investigation and analyses are compiled.
Conclusions and recommendations for the project implementation.

2. **What do you mean by “Pavement Distress”? Mention in tabular form, symptoms, causes and treatments of defects in bituminous surfacing of Nepal.**

Answer:

The visible signs of deterioration, damage, or failure that develop on a pavement surface due to traffic loading, weathering, poor construction, inadequate drainage, or aging is called pavement distress. These defects reduce riding quality, safety, structural capacity, and the service life of the pavement.

Defect	Symptoms	Causes	Treatment/Remedy
Cracking (Alligator/Fatigue, Longitudinal, Transverse cracking)	Interconnected cracks resembling crocodile skin	Caused due to repeated traffic loading, weak base or subgrade, temperature changes, aging of bitumen	Seal minor cracks, patch damaged areas, strengthen the base layer, provide overlay or reconstruction for severe cases
Rutting	Longitudinal depressions in wheel paths	Weak pavement layers, inadequate compaction, heavy axle loads, unstable bituminous mix	Remove rutted layer and replace with properly designed and compacted bituminous mix and strengthen pavement
Potholes	Bowl-shaped holes on pavement surface	Water infiltration, neglected cracks, traffic impact, poor drainage	Cut and clean pothole, apply tack coat, fill with hot/cold bituminous mix, compact properly.
Corrugation and Shoving	Wave-like ridges or bulges, especially near intersections and steep slopes	Unstable mix, inadequate compaction, braking and accelerating traffic	Remove affected area and replace with stable mix and improve compaction and mix design.
Depression/Settlement	Localized sunken areas retaining water	Weak subgrade, poor compaction, consolidation of foundation	Excavate failed portion, improve subgrade, reconstruct pavement layers
Stripping of bitumen	Separation of bitumen from aggregates	Moisture damage, poor aggregate-bitumen adhesion	Replace damaged mix, use of anti-stripping agents and ensure proper drainage

3. **Write differences between suspension and suspended trail bridge. Explain briefly the components that are to be covered in feasibility study of trail bridge.**

Answer:

The rugged and complex mountainous terrain of Nepal makes suspension bridges and suspended trail bridges an ideal solution for crossing rivers and deep valleys. The longest suspension bridge in Nepal is the Susta Suspension Bridge in West Nawalparasi, measuring 1,571 meters long and 1.6 meters wide.

The differences between suspension bridge and suspended trail bridge in design considerations and specific planning are listed below:

Suspension bridge	Suspended trail bridge
The bridge is designed to carry motor vehicles and heavy traffic.	This bridge is designed to carry pedestrian, livestock and cyclists.
They are of longer spans 100-200 m or more.	They are of short to medium spans 20-300 m.
Suspension bridge has wide deck of width 3.5-10 m or more for one or more traffic lanes.	Suspended trail bridge has narrow deck of 1-2 m suitable for pedestrian movement.
The design load considered in suspension bridges are heavy vehicular loads, impact loads, wind loads and seismic loads.	The design load considered in suspended trail bridge are pedestrian load, livestock load, wind and seismic loads.
The bridge requires strong and deep foundations.	The bridge requires comparatively lighter foundations and anchorages.
The deck has high stiffness that minimizes vibration and deflection.	The deck has moderate stiffness and limited deck movement.
The construction cost of suspension bridge is very high due to heavy structural components.	The construction cost of suspended trail bridge is comparatively low and economical.



a. Suspension bridge



b. Suspended trail bridge

Feasibility study of a trail bridge is carried out to determine whether the bridge is technically, economically, environmentally, and socially viable before detailed design and construction. The main components are as follows:

i. Assessment:

- The necessity of the bridge at particular location is identified and the existing crossing facilities are assessed.
- The population and settlements required to be served by the trail bridge is also determined, along with the evaluation of socio-economic benefits such as access to schools, health centres, markets etc.

ii. Site investigation:

- The most suitable bridge location is to be selected and river characteristics are studied.
- The topography and accessibility to the site is examined.

iii. Topographical survey:

- River banks surrounding terrain and trails are surveyed.
- Contour maps of the area and the longitudinal and cross sections are prepared.

- The bridge alignment is established and span length is estimated.
- iv. Geological and geotechnical investigation:
 - Soil and rock conditions are assessed at the foundation sites.
 - The bearing capacity of the soil is determined and foundation type is also selected.
 - Landslide prone areas, erosion and slope stability issues are identified.
- v. Hydrological investigation:
 - Design flood discharge is estimated and high flood level is determined.
 - River morphology, sediment transport and scour depth are assessed.
 - Seasonal flow variations in the river are also evaluated.
- vi. Technical feasibility:
 - The suitable bridge type is determined.
 - Bridge span, tower height, anchor block requirements and structural layout is estimated.
 - Availability of construction materials and equipment is also checked.
- vii. Environmental assessment:
 - Impacts on forests, wildlife, water quality and river ecosystem, is identified.
 - The effects during construction and operation are also assessed.
 - Mitigation measures and environment management plans are recommended.
- viii. Social assessment:
 - The impacts on local communities are assessed.
 - Land acquisition and resettlement needs are identified.
 - Community participation and willingness to contribute labor or resources are evaluated.
 - Benefits to women, children, elderly and disadvantaged groups are to be benefitted.
- ix. Economic and financial analysis:
 - Construction, operation and maintenance costs are estimated.
 - Cost-benefit analysis are conducted and economic viability is evaluated.
- x. Construction planning:
 - Accessibility to the site for transporting materials are assessed.
 - Construction methodology and sequence are identified.
 - Project duration, labor requirements are estimated.
 - Potential construction risks are identified.
- xi. Operation and maintenance planning:
 - The annual maintenance requirements and costs are estimated.
 - Responsibility is assigned for routine inspection and maintenance.

4. **Define break pressure tank. Explain the hydraulic specifics in pipeline network system planning and design keeping the minimum head loss in mind in this system.**

Answer:

Break pressure tank is a pressure releasing device to break the pressure with float valve to prevent bursting of pipe due to excessive pressure. It is provided in distribution mains.

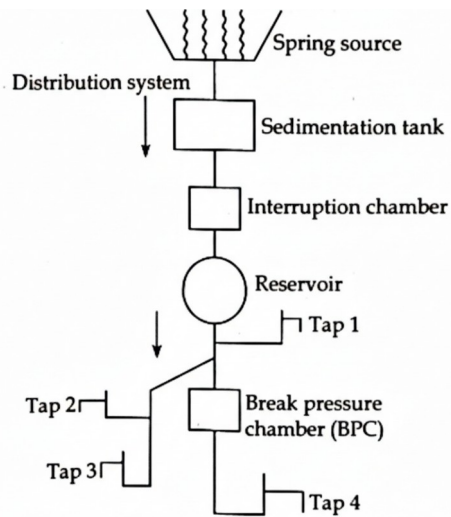


Figure: Typical rural water supply system

The hydraulic design of a pipeline network aims to deliver the required discharge with minimum head loss, adequate pressure, and economical pipe sizes.

- i. Proper selection of pipe diameter:
 - Adequate pipe diameters must be selected to carry the design discharge efficiently.
 - Larger diameters pipe reduces friction losses but increase construction costs.
 - Economic analysis plays an important role in pipe size selection.
- ii. Maintenance of optimum flow velocity:
 - The velocity recommended for flow in pipes are 0.6 m/s-2 m/s.
 - Very low velocity causes sediment deposition.
 - Excessively high velocity increases friction loss and water hammer.
- iii. Minimize friction loss:
 - The friction loss depends on length of pipe, diameter of pipe, flow velocity, pipe roughness and so on.
 - To minimize the fiction loss in pipe, smooth pipes with shorter length is preferred.
- iv. Reduction of minor loss:
 - Minor losses in the pipe occurs due to bends, elbows, valves, tees or sudden expansion or contractions in the pipe.
 - It can be minimized by using ling radius bends, limiting unnecessary fittings and providing gradual transitions.
- v. Alignment of pipeline:
 - The shortest practical route s adopted by avoiding steep rises and following the stable ground to reduce future maintenance.
- vi. Pressure zoning:
 - The system is divided into pressure zones using break pressure tanks where elevation differences are large.
 - Excessive pressures are prevented at lower elevations.
- vii. Air valve and scour valve provision:
 - Air release valves are installed at high points to remove the trapped air.
 - Scour valves are installed at the low points for flushing sediments.
- viii. Adequate Residual Pressure:

- Sufficient residual pressure must be maintained at the consumer tap, minimum of 7-10 m at the rural water supply system.
- Negative pressure conditions must be avoided.

ix. Looping of distribution network:

- Looped networks are preferred over dead-end systems where feasible.
- It ensures multiple flow paths and reduces the head loss.

x. Hydraulic analysis:

- Head loss is calculated using Darcy-Weisbach's formula or Hazen Williams equations.
- Hydraulic gradient line and residual pressure must be checked.
- The size of pipes must be optimized to achieve minimum head loss with economical cost.

5. **You are required to design a water supply scheme based on gravity flow system. Describe specific stage selection from sources, design of the system up to distribution of water supply to the community.**

Answer:

A gravity flow water supply system is a system in which water flows from a source at a higher elevation to the consumers at a lower elevation solely under the influence of gravity, without the use of pumps. The design process involves several stages, from source selection to water distribution.

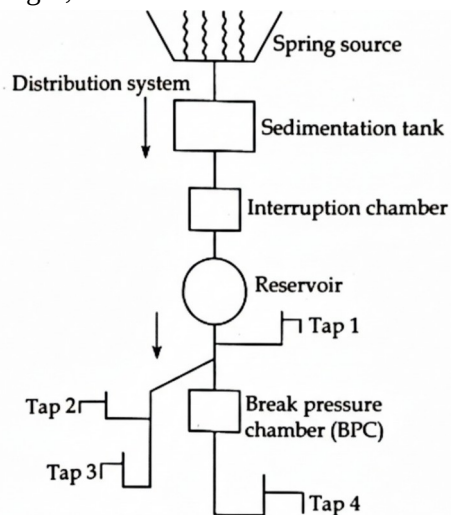


Figure: Typical rural water supply system

The stages in the design of gravity water supply systems are:

i. Source selection:

- Reliable source of water is selected with adequate supply of water at all times.
- It must have good quality of water at the higher elevation than the service area.
- It must be easily accessible for construction and maintenance.
- Some of the common sources of water are springs, rivers, streams, lakes, reservoirs, aquifer etc.

ii. Detailed investigation:

- Detailed investigations are conducted to determine the suitability of the source.
- Yield measurement, water quality testing, topographical survey, geological and geotechnical survey are conducted.

iii. Estimation of water demand:

- The estimation of the quantity of water required is based on present population, future population, design period, per capita demand, fire demand and so on.

iv. Design of intake structure:

- The intake structure draws the required amount of water from the source.

- It must prevent the entry of debris, silt into the hydraulic structure, protect against floods and soil erosion and ensures uninterrupted water collection.
- v. Design of transmission mains:
 - Transmission mains carry water from the source to the reservoir.
 - Flow velocity is to be maintained between 0.6-2 m/s in the pipe.
 - Durable pipe materials such as HDPE, DI, PVC pipes are used.
- vi. Water treatment:
 - The treatment process depends upon the quality of the water sources.
 - It includes: screening, plain sedimentation, sedimentation with coagulation, filtration, chlorination, aeration and so on.
- vii. Reservoir design:
 - A service reservoir stores the treated water and regulates the water supply.
 - It helps to balance the fluctuations between supply and demand and provide emergency storage.
 - The reservoir maintains adequate pressure in the distribution network.
- viii. Design of distribution system:
 - The distribution system delivers water from the reservoir to consumers.
 - The residual pressure of 7-10 m is maintained at the consumer taps.
 - Gate valves, air valves and scour valves must be installed and head losses are minimized.
- ix. Household connection and public standposts:
 - Water is supplied through individual house connections and public standposts for communities.
 - Public standposts should be easily accessible, hygienically maintained and conveniently located.
- x. Operation and maintenance:
 - It includes routine inspection of pipelines, leak detection and repair, community participation and so on.

6. **What is the importance of small hydro-project in Nepal? Describe briefly the elements that are to be considered in the design of such hydro power projects.**

Answer:

Small hydropower projects are those projects with a capacity between 1 MW and 25 MW. They require Initial Environment Assessment before construction. Some of the small hydropower projects of Nepal includes Sundarjal hydropower projects, mailing khola hydropower project, Sunkoshi hydropower project and so on. In Nepal's 2025 Department of Electricity Development (DoED) schedule for hydropower projects below 1 MW, there are 18 small/micro hydropower projects, with a combined capacity of 14.222 MW.

The importance of small hydro-power projects are as follows:

- i. It helps in rural electrification that provides electricity to remote and mountainous areas where national grid extension is difficult and costly.
- ii. Small hydropower project also promotes local industries, agriculture, tourism through reliable electricity supply.
- iii. It creates jobs during planning, construction, operation and maintenance.
- iv. It helps to diversify energy sources and contributes reliable electricity supply.
- v. Small hydro projects also play important role in local participation and improvement of infrastructures such as road, bridges etc.

The elements that are to be considered in the design of small hydropower projects include:

- i. Hydrological investigation:

- The discharges and seasonal flow variations in the river is measured.
 - Design discharge is estimated and dependable flow.
 - Flood frequency and sediment load is assessed.
- ii. Topographical survey:
- The catchment area and river alignment are surveyed.
 - Suitable location is selected for project components.
- iii. Geological and geotechnical investigation:
- Foundation conditions are assessed and landslide-prone and unstable areas are identified.
 - Rock and soil characteristics are also studied.
- iv. Selection of intake site:
- Sufficient water availability in the source must be ensured with minimum sediment entry.
 - It must be located at the place where it provides easy access for maintenance.
- v. Design of headworks:
- It includes the design of diversion weir, intake, gravel trap, de-sanding basin etc.
- vi. Water conveyance system:
- It includes the design of canal, forebay, penstock that conveys water from intake or reservoir to the turbine of hydropower project.
- vii. Design of penstock:
- The suitable diameter and material are selected.
 - Anchor blocks and supports for penstock is also designed.
- viii. Design of powerhouse and tailrace:
- Appropriate layout is selected for safe operation and maintenance.
 - Suitable type of turbine, generator and control equipment are also selected.
- ix. Environmental and social considerations:
- The impacts of project on the environment should be minimized.
 - Land acquisition and resettlement issues must be addressed.
- x. Economic analysis:
- The cost required for construction, operation and maintenance are estimated.
 - Cost-benefit analysis should be conducted to determine the financial viability of the hydropower projects.
- xi. Operation and maintenance:
- It includes routine inspection of mechanical and electrical equipment.
 - It also ensures the accessibility for inspection and repairs.

7. If you have to design an irrigation system to be totally maintained by local farmers after construction, what factors would you consider in deciding about the lined and unlined section of the canal? Give technical and economic reasons and discuss.

Answer:

The irrigation system that is designed to be totally maintained by local farmers after construction is widely known as Farmers Managed Irrigation System (FMIS). It must be durable, affordable, sustainable and easy to maintain. The canal can be lined or unlined depending upon technical, economic, social factors.

The technical factors to be considers in deciding about the lined and unlined section of the canal are given below:

- a. Soil characteristics:
 - Unlined canals are suitable where the soil is stable, clayey and less permeable because of less seepage loss.
 - In sandy, gravelly or highly permeable soils, lined canals are preferred to reduce excessive seepage and erosion.
- b. Seepage loss:
 - Canal lining should be provided where seepage losses are significant.
- c. Topography:
 - Lining is recommended in steep slope where there is high risk of soil erosion and instability.
 - Similarly in flat terrains with low velocity of flow, unlined canals can be provided.
- d. Flow velocity:
 - High velocity of water flow may cause erosion of the beds and banks of canals.
 - Hence, lining is provided in the canal where the flow velocity exceeds permissible velocity.
- e. Water availability:
 - Where water is scarce, lining helps to conserve water by minimizing seepage loss.
 - Unlined canals can be provided where availability of water is abundant.

The economic factors to be considered in deciding about the lined and unlined section of canals are given below:

- a. Initial cost of construction:
 - Initial cost of construction for the lined canals is high.
 - Unlined canals have low initial cost during construction.
- b. Operation and maintenance cost:
 - Operation and maintenance cost of lined canal is less as they are less damaged by water flow.
 - Unlined canals require continuous maintenance as they are affected by flow velocity.
- c. Benefit-cost analysis:
 - Lining of canals becomes economically viable when the water is scarce and large seepage occurs in the area.
 - Unlined canals are economical where the seepage losses are small and maintenance labor is readily available.

8. **A canal had bed width of 6 m, full supply depth 1.6m, bank width 2.5 m, cutting slope 1:1, filling slope 1.5:1 and free boards 0.4 m. Calculate the balancing depth.**

Answer:

Given,

Bed width, $B = 6\text{m}$

Full supply depth, $d = 1.6\text{m}$

Bank top width, $t = 2.5\text{m}$

Cutting side slope = 1:1

Filling side slope = 1.5:1

Freeboard = 0.4m

Hence,

Height of bank above bed

$$H = d + \text{freeboard} = 1.6 + 0.4 = 2.0 \text{ m}$$

Let balancing depth (depth of cutting below ground level) be x .

Then:

$$\text{Depth of cutting} = x$$

$$\text{Height of embankment} = 2 - x$$

Area of excavation:

$$A_c = Bx + m_c x^2$$

$$A_c = 6x + x^2$$

Area of one bank:

$$A_b = t(2-x) + \frac{m_c (2-x)^2}{2}$$

$$= 2.5(2-x) + \frac{1.5 (2-x)^2}{2}$$

Since there are **two banks**,

$$A_f = 2A_b$$

$$= 5(2-x) + 1.5(2-x)^2$$

Equating embankment and excavation:

$$6x + x^2 = 5(2-x) + 1.5(2-x)^2$$

Expanding,

$$6x + x^2 = 10 - 5x + 6 - 6x + 1.5x^2$$

$$6x + x^2 = 16 - 11x + 1.5x^2$$

$$0.5x^2 - 17x + 16 = 0$$

Multiplying by 2,

$$\begin{aligned} x^2 - 34x + 32 &= 0 \\ x &= \frac{34 \pm \sqrt{34^2 - 4(32)}}{2} \\ &= \frac{34 \pm \sqrt{1028}}{2} \\ &= \frac{34 \pm 32.06}{2} \end{aligned}$$

Possible roots:

- $x = 33.03 \text{ m}$ (not possible)
- $x = 0.97 \text{ m}$

Hence, Balancing depth $\approx 0.97 \text{ m}$

9. Discuss on techniques and technology adopted in traditional construction in Nepal to make earthquake resistant. Show the elements in sketches.

Answer:

Traditional Nepalese buildings are not designed using the modern engineering principles, but many indigenous construction techniques improve in structural stability, flexibility and resistance to seismic forces. In Nepal, we can see the use of timber bands in windows, plinth and gable, in many residential buildings rural Nepal.

The techniques and technologies adopted in traditional construction in Nepal to make earthquake resistant are as follows:

a. Timber bands:

- Horizontal timber bands are provided at plinth level, lintel level and roof level.
- These bands tie the walls together and distribute the earthquake forces uniformly.
- They reduce the chances of wall separation and collapse.



b. Timber frames with masonry infill:

- A wooden frame carries most of the structural load.
- Brick or stone masonry is used as infill.
- It provides flexibility and ductility.



c. Wooden pegs and traditional joints:

- Timber members are connected using wooden pegs and interlocking joints.
- The connections remain flexible and absorb the earthquake energy.
- It is less brittle than rigid connections.



d. Lightweight roof construction:

- Traditional roofs use timber trusses with slate or light roofing materials.
- Uniform load distribution minimizes torsional effects.
- Simple geometry performs the better during earthquakes.



e. Thick stone and brick masonry walls:

- Thick walls provide the stability against gravity loads.
- It is combined with timber bands, which performs better than unreinforced masonry alone.



10. New technological innovation has resulted into a lot of benefits to the society. Justify these statements with appropriate examples.

Answer:

The development and application of new materials, methods, equipment, and systems to improve efficiency, quality, safety, and sustainability can be said as technological innovation. In this 21st century innovations and technological advancements are the crucial part of society. These innovations are developed to help humans do different works easily and rapidly. Technological advancements in the construction sector, have significantly improved people's quality of life and accelerated economic development.

The benefits of technological innovations are listed below:

- i. Improved efficiency in construction:
 - Modern equipment and machinery reduce the construction time and labor requirements.
 - Mechanized construction increases productivity and accuracy.
 - Example: Excavators, tower cranes, mix concrete plants etc.
- ii. Earthquake resistant construction:
 - New innovation and technologies also help to strengthen structural safety in earthquake-prone regions like Nepal.
 - Example: RCC framed structures, seismic resistant detailing, base isolation, energy dissipation device.
- iii. Better quality and durability:
 - Advanced construction materials improve the strength and lifespan of the structures.
 - Example: high-strength concrete, Fiber Reinforced Polymer, geotextiles etc.
- iv. Sustainable and environment-friendly development:
 - New technology also reduces the environmental impacts and promotes the efficient use of resources.
 - Examples: rainwater harvesting systems, green roofs, wastewater recycling, energy efficient buildings etc.
- v. Better water supply and irrigation:
 - It helps to improve the water conservation and agricultural productivity.
 - Examples: Drip irrigation, sprinkler irrigation, SCADA systems integration for water supply management.
- vi. Better project planning and management:
 - Computer based tools improve project scheduling, cost estimation and quality control.
 - Examples: AutoCAD, Civil 3D, Primavera P6, Microsoft Project etc.

- vii. Modern survey and mapping:
 - Survey work has become more accurate and less time-consuming.
 - Examples: Total Station, GNSS surveying, LiDAR technology.
- viii. Economic growth and employment:
 - Technology promotes industrialization, creates skilled jobs and increase the national productivity.
 - It supports infrastructure development and attracts investment.